Student ID _____ Student Name _____

University of the South Pacific

Faculty of Science, Technology and Environment

School of Computing, Information and Mathematical Sciences

IS221 Web Applications Development

Mid Semester Test

Semester 2, 2015

Possible Marks:	**100**
Weight:	**15%**
Time Allowed:	**120** minutes
Number of Pages:	**12** (including this page)

Instructions:

1. Sections A, B, C, D : Answer ALL questions.

3. Write ALL answers in the spaces provided in the Answer Booklet.

- If you do need extra space for your answers, ask the lecturer.

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Section A: Multiple Choice 30 Marks

Instructions:
Identify and circle the choice that best completes the statement or answers the question. There is only **one** possible answer.

1) The _____ is an Internet –based distributed information system.
   a. Web
   b. Net
   c. Internet
   d. Server
2) _____ is a markup language that lets you identify common sections of a Web page.
   a. Internet
   b. HTML
   c. Tags
   d. Document
3) The World Wide Web links information on related topics using _____
   a. Hypertext
   b. Tags
   c. URLs
   d. URIs
4) In Lab4, Microsoft Visual Web Developer Express 2010 is an example of an HTML editor/web development environment that is both _____ and _____.
   a. Code based, WYSIWYG
   b. Code based, WYSINWYG
   c. Graphics based, WYSIWYG
   d. Graphics based, WYSINWYG
5) Migration from legacy HTML to HTML5 should be a _____ process.
   a. Compatible
   b. Gradual
   c. Prolonged
   d. Portable
6) Google searches make use of the _____ tags defined in the _____ section of an HTML document:
   a. <title>, header
   b. <meta>, header
   c. <meta>, body
   d. <title>, body
7)
 and <img> are examples of _____ elements
   a. Container
   b. Empty
   c. Inline
   d. Block
8) Deprecated elements such as <br> and <font> are allowed in which version/s of XHTML?
   a. Strict
   b. Transitional

c. Frameset
d. Both Frameset and Transitional

9) The _____, specifies the rules for the document language and forces the browser to display in standards mode
   a. Version
   b. Elements
   c. Doctype
   d. Head

10) <img alt="company logo" src="myphoto.jpg" width="300" height="200">, is an example of ensuring accessibility for the blind using the _____ attribute..
   a. alt
   b. src
   c. width
   d. height

11) A/An _____ stores elements and attributes unique to a DTD that could be looked up by the browser when rendering an XHTML document..
   a. Tag
   b. Namespace
   c. Declaration
   d. Class

12) XHTML does all the formatting at the _____ rather than at the _____
   a. server, browser
   b. browser, source
   c. style sheet, browser
   d. document, browser

13) (URLs) or Web addresses connect to a specific _____
   a. Database server
   b. Web server
   c. Email server
   d. File server

14) A Web site's design should _____ the content and _____ the reader
   a. support, compliment
   b. compliment, guide
   c. compliment, support
   d. guide, support

15) In planning for clear presentation, the designer should take into consideration that users do more _____ and less _____
   a. reading, scanning
   b. browsing, reading
   c. scanning, browsing
   d. scanning, reading

16) Placing navigation elements in the same position on each page is an example of planning for _____
   a. smooth transition
   b. accessibility

c. legibility
d. uniformity

17) Fixed, flexible and f-shaped are examples of how users traverse your page and should be considered when designing for _____
a. smooth transition,
b. location
c. legibility
d. accessibility

18) _____ white space is the result of mismatched shapes
a. Active
b. Passive
c. Linear
d. Uniform

19) Online feedback forms and Google analytics are examples of tools used to:
a. Know you users
b. Track your users
c. Identify your users
d. None of the above

20) In file naming conventions, this standard works across all operating systems
a. ISO 27037
b. ISO 9660
c. ISO 27000
d. ISO 2400

21) _____ URLs are the unique address of a file on the Web
a. Complete
b. Partial
c. Relative
d. Absolute

22) In lab 4, you created an Images folder and placed all you pictures in it. This is an example of _____ folder.
a. hierarchical
b. single
c. flat
d. relative

23) In _____ design, the designer controls the view of the content.
a. Flexible
b. Fixed-width
c. Relative
d. Absolute

24) Domain names are managed by _____
a. W3C
b. ICANN
c. CISCO
d. IEEE

25) Style rules are composed of two parts: a _____ and a _____
   a. Selector, declaration
   b. Attribute, attribute value
   c. Selector, attribute
   d. Declaration, attribute
26) In CSS, _____ style only applies to the document it is declared in
   a. Internal
   b. External
   c. Inline
   d. outline
27) .css is an indicator of _____ styles
   a. External
   b. Internal
   c. Inline
   d. Outline
28) Class selectors are indicated using the _____ flag character
   a. .
   b. #
   c. :
   d. *
29) p {font-family: arial, helvetica, sans-serif;} is how you can specify _____ fonts
   a. absolute
   b. alternative
   c. downloadable
   d. relative
30) The user's _____ and _____ determine how a font is displayed
   a. Device, browser
   b. Browser, operating system
   c. Device, operating system
   d. Browser, resolution

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Section B: Matching Questions 20 Marks

Write the letter of the correct term that matches the given description in the space provided

A. *W3C*

B. *HTML5*

C. *Universal selector*

D. *Maintainability*

E. *Structure*

F. *Type selector*

G. *
*

H. *Inline*

I. *Browser cache*

J. *Core Web fonts*

K. *Definition lists*

L <?xml version = "1.0"?>

M *Div*

N *Hyperlink*

O *Operability*

P *Domain name*

1. ___B_____ All display elements have been removed in favor of CSS

2. ___A_____ Sets standards for HTML and other markup languages.

3. ____L____Tell the browser which document version you are using .

4. _____C___ * {color: purple;}

5. ____D____.Advantage of using CSS

6. ___T_____ Information and user interface components must be perceivable by users

7. ___H_____ Enables parts of text content to be treated differently from the surrounding text

8. _____F___ H1 {color: red:}

9. ____E____ Parts of the page such as the *title*, a *heading*, a *block* such as a *division* or *paragraph*, or a *cell* in a table

10. ___J_____ Set of fonts designed for specific use on the Web.

11. ____I____ On return visits, your computer loads the files locally unless content has changed

12 ____G____ Break up a line at a certain point

13 _____Q___ Renders text in bold.

14 ___M_____Used to identify blocks which will have common style properties

15 _____K___ Use **<dt>** and **<dd>** instead of **<li>**

Q **

R **

S *Grouping selectors*

T. *Perceivability*

16 ____P____ An alias that points to your actual location on the Web server

17 ___R_____ *indicate a section of text that the author wants to identify as significant.*

18 ____S____ h1, h2 {color: red;}

19 ___N_____ <a href="laptop.htm">laptop</a>

20 ___O_____ User interface components must be operable by users

Section C: Coding Questions 20 Marks

1) Given the source code below, with the help of a diagram illustrate how the web page is going to be rendered. **For <title> insert your name e.g. Vara's World** (10 Marks)

```
<!doctype html>
<HTML>
<head>
<Title>Your name World</Title>
<style type="text/css">
</style>
</head>
<Body background="background.jpg">

<Center>
<font size="10"
<h1 style="Color:blue"><b>Welcome to my cyber Home<b></h1>
</font>
<BR/>
<font size="5"
<h2>My second home ☺ </h2>
</font>
<BR/>
<BR/>
<font size="8"
<hr color="DarkBlue" size=10px>
<font size="5"
<p><b>Menu of Options</b></p>
</font>

</center>


<ul {margin-left: 50px;} >

<i>
```

```
    <li><a href="http://www.Facebook.com/">Where I catch up with my
friends</a>
    <li><a href="http://www.youtube.com">Where I get most of my music</a>
    <li><a href="http://www.twitter.com">Where I whine about life</a>
    <li><a href="http://moodle.usp.ac.fj/">My favorite class</a>
    <li><a href="http://webmail.student.usp.ac.fj/">Student Mail</a>
    </i>
</ul>

<p><i><b>
Do keep me posted ☺
<a
href="mailto:Your_Student_ID@student.usp.ac.fj">Your_Student_ID@student.usp.a
c.fj</a>
</b></i></p>

<hr color="DarkBlue" size=10px>
</body>
</html>
```

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2) Given the source code below identify five specific errors, state the XHTML syntax rule that the error violates and state how to correct it using examples. (10 Marks)

Error 3

```
<!DOCTYPE html
PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
```

Error 1

```
<HTML>
<head>
   <title>Table of Contents</title>
</head>
<body>
```

Error 2

```
   <h1>
      <font face=sans-serif color="#993333" align=center>
         Introduction to HTML &minus; Table of Contents
      </font>
   </h1>
   <ol>
     <li>
      <dl>
         <dt><b>Introduction to the HTML</dt></b>
         <dd> Basic HTML Elements, CSS &alt; XHTML<br/></dd>
      </dl>
     <li>
      <dl><b>Body of an HTML Document</b>
      <li type=a>
         Organization of an HTML Document </dd><br />
         <dd> Lists</dd><br />
         <dd>   <ol type="square">
                        <li>DL (Glossary) Lists</li>
                  <li>Regular (UL, OL) Lists<br /></li>
            </ol>
         </dd>
          </dl>
   </li>

   <li>
      <a HREF="test">Bibliography and References</a>
      <br>
   </li>
   </ol>

   <hr width="85%" />

</body>
</HTML>
```

Error 4

Error 5

Error 1_____<HTML> - Element name in Capitals_____

Syntax rule violated ___All elements should be in lower case_____

Correction_____<html>_____

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Error 2_____<font face=sans-serif color="#993333" align=center>_____

Syntax rule violated _____All attribute values should be in quotation marks___

Correction_____<font-face="sans-serif align="center">_____

Error 3_____Missing document version in doctype declaration_____

Syntax rule violated _____Document declaration includes document version, document type and namespace_____

Correction___Add document version_____

Error 4___<dt>Introduction to the HTML</dt></b_____

Syntax rule violated _____All elements should be properly nested_____

Correction___<dt>Introduction to the HTML</dt> _____

Error 5___as depicted _____

Syntax rule violated _____Document must be well formed_____

Correction_____Properly nest elements and use <dl> and <dd> tags instead of <li>____

Section D: Short Answer Questions 30 Marks

1) You have been asked to develop a website for a company that is proposing to rent apartments for USP student. Using the **web development lifecycle** discuss how you would go about implementing this project. Your discussion should include the **key stages of the cycle** and examples of **activities** and **artefacts** involved in that stage. (10 Marks)

Requirements specification

Activities

- Determine how the customer wants to use the site e.g. will there be credit card payments etc. Will there be a catalogue for customers to view and select the kava packs they want. How about delivery, will you customers need to provide their delivery address etc.

- Type of website in this case will be an e-commerce website therefore think of common elements that an e-commerce website should have.
- Since it's going to be an e-commerce website you will need to think of an appropriate domain name e.g. renting online a .com domain would be most appropriate.
- Confirm requirements and specifications with client.

Artefacts

- Registered domain name (www.uniroomsforrent.com.fj)
- Requirements specifications document which will consist of:
  - Technical specifications e.g. online searchable catalogue, form to enter customer's details etc and a database to store catalogue details and if need be customers' details.
  - *Design requirements* e.g. colour scheme or a theme to be used for the website that is in line with the Uni rooms for rent concept
  - *Page layout sketches* e.g. a sketch of what the site may look like
  - *Audience definition* e.g. who your potential customers are, will they be just local customers or international customers

Information design and taxonomy creation

` Activities

- Sorting out the web pages the customer wants for instance what content to have on what page and how to structure content and hyperlinks etc. Think in terms of the layouts discussed in readings, customer might want to have a featured product/rental specials or something
- Sketch of what the pages and the website may look like and navigation between pages etc.
-

Artefacts

- A sketch of the various web pages and how they should be interlinked based on their content

Graphic design and page template creation

Activities

- Creating web graphics, and page templates e.g. logo for Uni rentals should be representative of their product and what the company wants to portray etc.
- Navigation buttons/scheme etc.
- Wireframe sketches of web site/web pages

Artifacts5

- Web graphics, web page templates

Construction and Web development/Quality Assurance and User Testing

Activities

- Coding of the required elements and content
- Construction of databases
- Testing of coded elements to ensure they are functioning correctly e.g. testing the room booking functionality etc.

Artefacts

- A database driven functioning website with forms, catalogues and components that are tested and functional.

Publishing and promotion

Activities

- Determining whether to host the website onsite (your own servers etc) or use a web hosting company.
- Web site is uploaded to web server and is now available and live
- Use Facebook for business to promote website or twitter etc

Artefacts

- Fully functional website

Maintenance

Activities

- Updating the website e.g. adding new products, on sale items etc.

Artefact

- Well maintained site that is available, up to date and fully functional

2) Discuss how you are able to view your Facebook page on your mobile phone. Your discussion should include the **core elements of web applications architecture** (10 Marks)

Client using HTTP/HTTPS protocol requests the Facebook login page. This request is put in an IP envelope and send over the Network. At Facebook's end, the IP envelope is opened and the request was send to one of FB's webservers. The webserver then interacts with the database server to retrieve the requested Facebook member's login page. The database server response with the requested login page to the webser, the webserver then packages this into an IP envelope and sends the response back onto the network to be delivdered to the requesting client.

3) Discuss how you would design for both Accessibility and Legibility (10 Marks)

Accessibility

Information and user interface should be perceivable to the users

User interface components should be operable by users

Information about the user interface its operations must be understandable by users

Content shuld be robust enough to be interpreted by wide variety of agents.

Legibility

Choice of fonts should be easy to read and universally available for users' browsers.

Choice of colour should complement content

Use graphics sparingly for emphasis

Use of active whitespace for structure and visual enhancement of text